Bo Liu

Tel.: +86-13973112505
Email: liubo317@hnu.edu.cn , Personal Website

Research Interests: Agentic AI, Spatial Intelligence, Vision-Language Models (VLMs)



EDUCATION

College of Computer Science and Electronic Engineering, Hunan University — Changsha, China
Master's Degree in Software Engineering — September 2024 -- June 2027

- Main Courses: Data Mining, Intelligent Computing, Advanced Database Systems, Advanced Software Architecture.
- Cumulative GPA: 88.80/100.

Maynooth International Engineering College, Fuzhou University — Fuzhou, China
Bachelor's degree in Robotics & Intelligent Devices — September 2020 -- June 2024

- Main Courses: Control Systems Design, System Dynamics, Computer Vision, Autonomous Mobile Robotics.
- Cumulative GPA: 3.84/4 (Top 3).

SELECTED RESEARCH EXPERIENCE

Multi-Agent Reasoning over Evolving User-Location Graphs for Human Mobility Generation — Hunan University
Graduate Student — May 2026 -- August 2026

- Proposed a multi-agent framework that reformulates human mobility as the evolution of user–location interaction graphs, extending mobility modeling from individual trajectories and population flows to community structures.
- Designed User, Location, and Manager Agents to reason from individual and community perspectives, coordinating individual intentions with mobility motifs, population dynamics, and community-level interaction patterns.
- Developed community-consistency reflection to diagnose structural deviations and selectively revise agent decisions.

Retrieval-Augmented Generation and Personalized Agents for App Usage Prediction — Hunan University
Graduate Student — November 2025 -- February 2026

- Proposed a Center-User multi-agent architecture that bridges collective and personalized patterns via collaboration.
- Developed a Collective Pattern Builder on the cloud center-side to construct a behavioral knowledge base and retrieve shared usage patterns, and a Personalized Pattern Profiler on the local user-side to maintain individual memory.
- Introduced an adaptive reflection mechanism for integrating behavior patterns and optimizing agent coordination.

LLM-Based Semantic Reasoning for Spatiotemporal Human Behavior Modeling — Hunan University
Graduate Student — April 2025 -- October 2025

- Investigated LLMs as semantic reasoning engines for modeling spatiotemporal human behavior dependencies.
- Developed an LLM-based unified mobility generation framework that infers high-level travel intent and guides cross-city trajectory generation through unified spatial representations and Diffusion models.
- Developed an LLM-based hierarchical app usage prediction framework that conducts long-term behavior compression, interprets contextual information, and performs coarse-to-fine reasoning.

Graph Learning and Diffusion Modeling for Spatiotemporal Mobility Prediction — Hunan University
Graduate Student — November 2024 -- February 2025

- Developed a multi-granularity self-guiding graph diffusion model for spatiotemporal private car activity prediction.
- Established a technical foundation in spatiotemporal intelligence spanning graph modeling, contrastive learning, attention-based feature fusion, and diffusion-based prediction.

PUBLICATIONS

- **Bo Liu,** Tong Li (Mentor), Jiapeng Zhang, Zhu Xiao (Supervisor), et al. Beyond Trajectory Statistics: Community-Consistent Human Mobility Generation via Evolving User–Location Graphs. *Proceedings of the ACM Web Conference 2027*. (Under Review)
- **Bo Liu,** Tong Li (Mentor), Miao Xiao, Zhu Xiao (Supervisor) et al. AgentApp: Bridging Personalized and Collective Patterns for Mobile App Usage Prediction via Collaborative LLM Agents. *The Forty-First AAAI Conference on Artificial Intelligence*, 2027. (Under Review)
- **Bo Liu**, Tong Li (Mentor), Miao Xiao, Zhu Xiao (Supervisor), *et al*. LLMApp: Unleashing the Power of Large Language Model for Mobile App Usage Prediction. *IEEE Transactions on Services Computing*, 2026. (Published)
- **Bo Liu**, Tong Li (Mentor), Zhu Xiao (Supervisor), *et al*. All Cities are Equal: A Unified Human Mobility Generation Model Enabled by LLMs. *IEEE Transactions on Knowledge and Data Engineering*, 2026. (Minor Revision)
- **Bo Liu**, Tong Li (Mentor), Zhu Xiao (Supervisor), *et al*. A Mobility Prediction Method Based on Retrieval-Augmented Generation and Gravity Model. *Chinese Journal of Computers*, 2026. (Accepted)
- **Bo Liu**, Tong Li, Zhu Xiao (Supervisor), *et al*. A Multi-Granularity Self-Guiding Graph Diffusion Model for Predicting Private Car Activity. *IEEE Transactions on Intelligent Transportation Systems*, 2026, 27(6):7172-7186. (Published)
- Zhu Xiao (Supervisor), **Bo Liu**, Linshan Wu, *et al*. Exploring Spatio-Temporal Carbon Emission Across Passenger Car Trajectory Data. *IEEE Transactions on Intelligent Transportation Systems*, 2025, 26(2):1812-1825. (Published)

HONOURS

- National Scholarship for Graduate Students, 2024 – 2025, 2025 -- 2026.
- First Prize Scholarship of Hunan University, 2024 -- 2025, 2025 -- 2026, 2026 – 2027.
- Excellent Undergraduate Student of Fuzhou University, 2023 -- 2024
- Undergraduate Students Scholarship of Fuzhou University, 2020 -- 2021, 2021 -- 2022, 2022 -- 2023, 2023 -- 2024.

PERSONAL SKILLS

- Programming: Python, Java, MATLAB, C, C#, SpringBoot, SQL, LaTeX.
- Languages: Mandarin (native), English (IELTS 6.5).

INTERESTS

Electronic Keyboard, Hiking, Cycling, Basketball.